



Department of Computer Science

EE-229 – Computer Organization and Assembly Language

FALL 2025

Instructor Name: Aleena Ahmad

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Office Location/Number: New Building (office # 64)

Office Hours: Mon, Wed 01:00pm-2:20pm

TA Name (if any): Abdullah, Abdul, Hammad

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Office Location/Number:

Office Hours:

Course Information

Program: BS

Credit Hours: 3

Type: Core

Pre-requisites (if any): DLD

Class Meeting Time: Section 3C 02:30am (Mon, Wed), Section 3A 01:00pm (Tues, Thurs), Section 3B 02:30pm (Tues, Thurs)

Class Venue: F-308 (3C), C-4 (3B), F-202 (3A)

Program Learning Outcomes

PLO#	PLO Name	Description
2	Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
4	Design/Development of Solutions	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

Course Learning Outcomes:

Course Learning Outcomes (CLOs)	BT Level	PLO Mapping
1. Demonstrate the basic concepts of computer organization including CPU, memories, and input/output and explain their purposes and interactions.	C2	2
2. Describe the working of important x86 assembly primitives, including arithmetic, branching, bit manipulation, addressing modes and interrupt handling.	C2	2
3. Apply the knowledge of Intel x86 architecture to develop moderately complex and well-modularized assembly programs.	C3	4

Course Textbook

- Assembly Language Programming Lecture Notes by Bilal Hashmi (BH).
- Assembly Language for x86 Processors Seventh Edition Kip R. Irvine (KI)
- Computer Organization and Architecture Designing for Performance Tenth Edition by William Stallings (WS)

Tentative Lecture Plan

Topics to be covered	#Lectures
Introduction to Computer Organization and Assembly language	0.5
Computer functions and Interconnection	0.5
Intro to intel architecture (registers, bus and memory) Getting started in assembly language	2
Data Transfer and Addressing Modes	2
Instruction set with examples and integer arithmetic	5
Procedures and stack	4
Display memory and string processing	5
Interrupts	4
Advanced Concepts of Assembly	5

(Tentative) Grading Criteria

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|----------------------|----|
| 1. Quizzes | 10 |
| 2. Midterms | 30 |
| 3. Final | 45 |
| 4. Assignments+ Proj | 15 |

Course Policies

1. Quizzes may be un-announced.
2. No makeup for missed quiz or assignment.
3. 80% attendance
4. 50% passing marks

Academic Integrity

- Plagiarism and Cheating against academic integrity. Both parties involved in such cases will face strict penalty (negative marking, F grade, DC)
- CODE/ ASSIGNMENT SHARING is strictly prohibited.
- Keep in mind that by sharing your code/assignment you are not helping anyone rather hindering the learning process or the other person.
- No excuse will be entertained if your work is stolen or lost. To avoid such incidents
 - Keep back up of your code on safe online storage, such as Google Drive, Drop box or One drive.
 - Do not leave your work on university lab computer, transfer your work to online storage and delete from the university lab computer (empty recycle bin as well)